Gender or History of Anxiety Disorder Diagnosis: Which Factors May Contribute to Procrastination Patterns in People with Anxiety?

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Abstract

This paper examines the association between procrastinatory behaviours and symptoms of anxiety among university students and then divides the overall sample by gender and history of anxiety diagnosis. The data were collected using an online questionnaire combining Penn State Worry Questionnaire (PSWQ) and Lay's Procrastination Scale. Upon gathering the answers from 56 adults, the results of statistical analyses showed that procrastinatory behaviour can be associated with anxiety, females tend to worry and procrastinate insignificantly more than males, and recorded anxiety diagnosis may contribute to higher levels of worry but not procrastination.

Keywords: generalised anxiety disorder; GAD; anxiety; procrastination; PSWQ;

Introduction

Procrastination is a widespread phenomenon across age groups and contexts, including academic, occupational, and personal fields. According to Ibili (2025), everyone procrastinates to some extent. Research consistently indicates that procrastination can impact daily life. Examples of this impact can be seen significantly in tasks such as daily functioning, goal completion, and task accomplishment. Apart from the impact on productivity, chronic procrastination is linked to emotional distress, higher levels of stress, and negative impact on quality of life (Gautam et al., 2019), exemplifying the clear importance as a psychological and high behavioural concern.

To understand, why individuals tend to procrastinate, research has focused primarily on internal psychological factors that contribute to avoidant behaviour. Anxiety was associated with excessive feelings of uncertainty, fear of inadequacy or failure, and the struggle to initiate tasks (Dugas et al., 2005) Those experiencing heightened levels of anxiety may delay tasks as a way of seeking short-term relief, progressing into long-term procrastination patterns over time. Generalised anxiety disorder (GAD), which is categorised by constant and excessive worry across different life domains, may then play a significant role in influencing procrastination-based behaviours.

GAD involves chronic worry, cognitive reflection, and heightened emotional distress; all of which can interfere with effective task management. The overlap between anxiety-related avoidance and procrastination suggests a possible relationship between the two reactions. Clarifying how signs of generalised anxiety relate to procrastination across academic, task-oriented, and personal contexts and understanding this relationship may provide insight into the mechanisms that sustain procrastination and inform more effective interventions.

Therefore, it was hypothesised that individuals reporting increased levels of generalised anxiety would also report greater levels of procrastination. Specifically, a positive correlation was expected between Penn State Worry Questionnaire (PSWQ) scores and scores on Lay's Procrastination Scale (H_0).

Regardless of the interest in the connection between the two, cultural beliefs, social norms, and clinical knowledge suggest that certain factors are contributing to this relationship. That is, females are often considered to be more "anxious" than males, and previously diagnosed anxiety disorder may manifest in the future.

Additionally, it was therefore hypothesised that the female group of the sample will show higher levels of anxiety than the male group, and, with regards to the initial hypothesis, that procrastination levels of the former will be higher than those of the latter (H_1). Similarly, people previously diagnosed with anxiety disorder were expected to exhibit higher levels of both worry and procrastination than those without such diagnosis (H_2).

[1] Was responsible for writing Methods and Results, designing the questionnaire, running the statistical analyses, and contributed equally with the second author in Conclusion.
[2] Was responsible for writing Abstract, Introduction, and Discussion, collected and finalised References, and edited the text ensuring grammatical correctness.

Methods

Participants

A total of 56 participants were recruited using convenience sampling, with several inclusion and exclusion criteria applied, and took part in the study voluntarily. The overall sample included insignificantly more females (57.1%) than males (42.9%).

Inclusion criteria required participants to be (1) between 18 and 30 years of age, (2) able to understand written English, (3) willing and able to consent and participate, and, (4) if diagnosed with anxiety or fear-related disorder(s) using ICD-11 criteria (e.g., generalised anxiety disorder, social anxiety disorder, separation anxiety disorder, selective mutism, panic disorder, agoraphobia, or any specific phobias) (World Health Organization [WHO], 2019), undergoing stable medical treatment. Exclusion criteria ruled out participants who (1) had a history of substance abuse, (2) were diagnosed with disorder(s) due to substance use or addictive behaviours or (3) with any other disorder(s) whose symptom(s) overlap or coincide with symptom(s) of anxiety or fear-related disorders (e.g., intrusive thoughts for obsessive-compulsive or related disorders) or can be interpreted as procrastination (e.g., depressive episode(s) for depressive disorders or bipolar disorder) (WHO, 2019), (4) underwent recent medication changes, (5) had severe cognitive impairments, or (6) reported recent self-harm attempts or suicidal thoughts.

Additionally, two predominant age groups were defined within the sample: nineteen-year-olds and people over 25 years of age (28.6% each). The majority of the respondents were pursuing a bachelor's degree while completing the survey or had completed this degree at the latest (58.1%).

Moreover, a large part of participants was not previously diagnosed with anxiety or fear-related disorders (75%), with exactly one individual (1.8%) who preferred not to disclose this information.

Procedure

The survey was conducted online using a self-report questionnaire. After accessing the link, participants were asked to read and agree with an informed consent form outlining the purpose of the study, the voluntary nature of participation, the anonymity of responses, and the right to withdraw at any point without negative consequences. Participation was anonymous, and no personal information (e.g., ID, names, e-mail addresses, or IP addresses) was collected. The survey required approximately 25 minutes to complete.

Measures

PSWQ, a self-report tool created specifically to assess the likelihood of excessive and uncontrollable worry, was used to measure anxiety-related symptoms (Wilson et al., 2025). The term "worry" is used by the authors of PSWQ to describe the primary symptom and defining characteristic of anxiety disorders rather than the diagnostic criteria (Meyer et al., 1990).

Therefore, rather than diagnosing anxiety disorders, the questionnaire in this study was used to evaluate individual differences in worry severity and frequency as anxiety-related symptoms. Internal consistency for the overall sample was α= 0.902.

Procrastination was measured using a self-report scale originally developed for student populations (Lay, 1986). Additionally, minor modifications in wording were made to several questions to improve language clarity while preserving their original meaning. Cronbach's α for the overall sample was $\alpha = 0.832$.

The scales consisted of 16 and 20 items, respectively. The participants were asked to decide whether each of 36 statements was uncharacteristic or characteristic of them using a Likert-type 5-point scale ranging from 1 = strongly disagree to 5 = strongly agree. Higher scores indicated higher levels of worry-related anxiety symptoms or procrastination. Several items were reverse-keyed according to standard scoring procedures to reduce response bias[3].

Data Analysis

Data were analysed using R (Version 4.5; R Core Team, 2025). Descriptive statistics (means, medians, standard deviations, minima, and maxima) were computed for all study variables. Scatter plots used for illustrating the relationships within the data and found results were created via jamovi (Version 2.6; 2025). Internal consistency was assessed using Cronbach's alpha coefficients. Pearson's correlation was conducted to examine the relationship between worry-related anxiety symptoms and procrastination. Statistical significance was evaluated at α = .05.

Design

This study used a quantitative, questionnaire-based design. It examined the association between procrastinatory behaviours and symptoms of anxiety among university students and then divided the overall sample by gender and history of their anxiety diagnosis.

Results

Descriptive statistics for the PSWQ the procrastination scale are presented in Table 1. Mean PSWQ scores show moderate levels of worry-related anxiety symptoms (N = 56, M = 3.24, SD = 0.78), whereas mean scores on the procrastination scale are slightly lower (N = 56, M = 2.92, SD = 0.63).

[3] For the list of questions used in a survey as well as the details about the reverse-keyed items see Appendix.

Table 1. Descriptive statistics for PSWQ and procrastination scales

	PSWQ	Procrastination
N	56	56
Mean	3.24	2.92
Median	3.34	2.95
Standard deviation	0.777	0.630
Minimum	1.19	1.55
Maximum	4.69	4.25

Pearson correlation analysis was conducted to identify the relationship between worry-related anxiety symptoms and procrastination. As shown in Table 2, the results demonstrated a moderate positive correlation between the two variables, $r(54) = .37$, $p = .005$.

Table 2. Correlation matrix for PSWQ and procrastination scales.

		PSWQ	Procrastination
PSWQ	Pearson's r	—	
	df	—	
	p-value	—	
	N	—	
Procrastination	Pearson's r	0.370**	—
	df	54	—
	p-value	.005	—
	N	56	—

Note. * p < .05, ** p < .01, *** p < .001

An illustration in the form of a scatter plot for the correlation matrix was created and is presented in Figure 1[4].

To explore gender differences in worry-related anxiety symptoms and procrastination, descriptive statistics were calculated separately for males and females. Table 3 and Figure 2[4] present these findings. The female part of the sample ($n = 32$) showed mean scores of 3.49 and 2.96, while the scores of the male part ($n = 24$) were 2.91 and 2.87.

Table 3. Descriptive statistics divided by gender.

	Gender	PSWQ	Procrastination
N	Female	32	32
	Male	24	24
Mean	Female	3.49	2.96
	Male	2.91	2.87
Median	Female	3.56	3.02
	Male	2.88	2.77
Standard deviation	Female	0.700	0.681
	Male	0.761	0.565
Minimum	Female	1.94	1.55
	Male	1.19	1.75
Maximum	Female	4.69	4.20
	Male	4.13	4.25

[4] For the figures illustrating the reported data see Appendix.

The analysis that divided the overall sample by the previous anxiety diagnosis demonstrated that the mean score on PSWQ of participants with previous anxiety diagnosis ($n = 13$) was 3.64, whereas the group without thereof ($n = 42$) showed a mean of 3.12. Procrastination scale mean scores of the two groups showed results of 2.88 and 2.94, respectively. Table 4 shows descriptive statistics for this analysis, while Figure 3[4] illustrates these findings.

Table 4. Descriptive statistics divided by history of anxiety diagnosis.

	Diagnosis	PSWQ	Procrastination
N	Yes	13	13
	No	42	42
Mean	Yes	3.64	2.88
	No	3.12	2.94
Median	Yes	3.50	2.75
	No	3.19	3.02
Standard deviation	Yes	0.702	0.668
	No	0.774	0.632
Minimum	Yes	2.56	1.95
	No	1.19	1.55
Maximum	Yes	4.69	4.25
	No	4.19	4.20

Discussion

The computed mean value for PSWQ corresponds to moderate but statistically significant levels of anxiety. The mean calculated for the procrastination test suggests that procrastination levels are still defined, albeit at a lower level. A moderate positive correlation between the results of the PSWQ and procrastination scale, reported previously, suggests that higher levels of worry can be associated with higher levels of procrastination. This supports the hypothesis that worry-related anxiety symptoms are related to procrastinatory behaviours (H_0).

When analysing demographic differences, the results showed considerably higher levels of anxiety in females than in males and marginally higher levels of procrastination in comparison between the two groups. The observed differences in procrastination scores between females and males did not reach any statistical significance. Therefore, the corresponding hypothesis (H_1) is only partially supported by the data. While females in the sample did experience higher levels of anxiety symptoms, this did not contribute to a significant chronic increase in procrastination symptoms compared to their male peers.

However, a higher mean value on PSWQ was demonstrated by the group of participants previously diagnosed with anxiety-related disorders and behaviours compared to the group without a history of diagnosis. This clinical history did not correlate with high levels of procrastination, which informs us to partially reject hypothesis (H_2), since no higher levels of procrastination were found in the former group. The history of anxiety-related diagnosis does predict a heightened level of cognitive worry over time, yet it does not imply that an individual will default to behavioural task avoidance more than non-clinical participants.

Possible Explanations

When analysing whether there is a correlation between gender and previous diagnosis of GAD has a significant role in predicting patterns within procrastinating behaviours, several psychological mechanisms need to be considered.

First, some of the core symptoms of anxiety are excessive worry, fear, and heightened threat perception. Engaging in daily or regular tasks may be perceived as highly threatening or overwhelming. The individuals with a history of GAD may be more likely to delay or procrastinate the start of a task, which is a maladaptive strategy to temporarily delay and reduce feelings of high distress (Borkovec, 2004). This may differ based on gender, as proven by empirical evidence showing there are distinct differences in how men and women react to and manage perceived threats.

Similarly, intolerance of uncertainty heavily drives these patterns (Carleton et al., 2012). There are many tasks that involve ambiguous outcomes or unrealistic timeframes; therefore, those with a long-standing diagnostic history may actively postpone them to avoid uncertainty.

Third, according to the attentional control theory, described by Eysenck et al. (2007), consistent worry causes severe disruption to goal-oriented behaviour, thereby triggering procrastination even when motivation is categorised as high (Rezaei-Gazki et al., 2024). The disruption is intensified by the feelings of perfectionism and fear of failure, which are widely observed in cases of clinical anxiety. The high personal standards and fear of making errors or mistakes typically result in neglect or avoidance (Gadosey, 2023; Steel, 2007).

Gender can be seen as a critical moderator; research suggests that there are many variables showcasing the differences between men and women regarding socially prescribed perfectionism and perception of being evaluated negatively (Wei et al., 2025). This alters the intensity with which one procrastinates.

Finally, the variables likely feed into the self-perpetuating cycles. Procrastination ultimately intensifies the symptoms of GAD by compounding time pressures, heightening feelings of uncertainty, and fuelling levels of self-criticism. Instead of fostering positive coping mechanisms, the cycle is reinforced by continued levels of anxiety while maintaining higher levels of stress over time; this framework is deeply inherent in those with a chronic history and previous diagnosis in comparison to those experiencing acute anxiety (Jochmann et al., 2024).

Limitations

Several limitations of this study should be acknowledged. First, the sample size was relatively small (N = 56) and obtained through convenience sampling, which limits the generalisability of the findings beyond university-aged populations similar to the one examined. Furthermore, the subgroup sizes were notably unequal, particularly within the diagnosis-based comparison (n = 13 vs. n = 42), which may have reduced statistical power and affected the reliability of results belonging to H_2.

Second, the cross-sectional and correlational nature of the design does not allow for any causal conclusions to be drawn regarding the relationship between worry-related anxiety symptoms and procrastination. While a moderate positive correlation was found, it remains unclear whether anxiety contributes to procrastination, whether procrastination exacerbates anxiety, or whether both are influenced by shared underlying factors. Thus, further research is needed.

Third, all data was collected via self-report measures, which are susceptible to social desirability bias and inaccuracies in self-perception. This applies not only to the PSWQ and procrastination scale but also to participants' reported history of anxiety diagnosis, which was not clinically verified. Consequently, although the PSWQ was used to assess levels of worry as a core feature of generalised anxiety, it does not constitute a diagnostic tool, and therefore conclusions regarding GAD specifically should be interpreted with caution.

Finally, minor wording modifications were made to Lay's (1986) procrastination scale to improve clarity, which, despite acceptable internal consistency ($\alpha = 0.832$), may limit direct comparability with findings from studies using the original measure.

Conclusion

This study investigated whether an association between worry-related anxiety symptoms and procrastination can be determined. After examining the relationship between the listed variables via a standardised self-report questionnaire, the findings revealed a moderate positive correlation between anxiety-related worry and procrastination behaviours. Hence, the initial hypothesis (H_0) was confirmed.

Furthermore, the results of further statistical analyses demonstrated that females tend to worry and therefore procrastinate more than males, although the latter result was considered insignificant. Yet, while people that reported previous anxiety disorder diagnosis were confirmed to worry more than people without thereof, higher procrastination scores of the former were not found. Thus, both the first and the second hypotheses (H_1 and H_2, respectively) were not entirely supported.

Additionally, the findings show that procrastination should not be viewed in isolation; instead, described behaviours should be considered as part of a psychological pattern associated with anxiety-related symptoms. Our data confirms that as levels of worry increase, so does task delay. The observed correlation suggests that procrastination may arise in the context of persistent worry, later resulting in challenges in task engagement when anxiety-related processes consume cognitive and emotional resources.

Overall, this study outlines the importance of interpreting procrastination by recognising the impact of anxiety-related symptoms. This paper shows empirical evidence of a significant correlation between the symptoms of GAD and procrastination, while confirming the importance of considering emotional and cognitive factors for its underlying mechanisms.

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Appendix

List of Abbreviations

GAD	Generalised Anxiety Disorder

ICD-11	International Classification of Diseases (11th Revision)
ID	Identity Document
IP (address)	Internet Protocol
PSWQ	Penn State Worry Questionnaire
SD	Standard Deviation
WHO	World Health Organisation

List of Questions Used in a Survey

Part I: Penn State Worry Questionnaire

1. If I do not have enough time to do everything, I do not worry about it.
2. My worries overwhelm me.
3. I do not tend to worry about things.
4. Many situations make me worry.
5. I know I should not worry about things; but I just cannot help it.
6. When I am under pressure I worry a lot.
7. I am always worrying about something.
8. I find it easy to dismiss worrisome thoughts.
9. As soon as I finish one task, I start to worry about everything else I have to do.
10. I never worry about anything.
11. When there is nothing more I can do about a concern, I do not worry about it anymore.
12. I have been a worrier all my life.
13. I notice that I have been worrying about things.
14. Once I start worrying; I cannot stop.
15. I worry all the time.
16. I worry about projects until they are all done.

Part 2: Lay's Procrastination Scale

17. I often find myself doing tasks that I had intended to do days ago.
18. I do not complete assignments until the deadline comes.
19. When I finish reading a library book, I immediately return it, regardless of the deadline.
20. When it is time to get up in the morning, I usually get out of bed right away.
21. A letter or message may be already written for several days before I send it.
22. I usually answer phone calls, texts, or emails quickly.
23. I can put off completing tasks for days, even those that require nothing more than sitting down and doing them.
24. I usually make decisions as soon as possible.
25. I usually linger before starting a task I have to do.
26. I usually have to rush to complete a task on time.
27. When I am getting ready to go out, I rarely get caught doing something at the last minute.
28. Even if I know about some deadlines, I often waste time doing other things.
29. I prefer to leave early for an appointment.
30. I usually start an assignment shortly after it is assigned.
31. I often have a task finished sooner than necessary.
32. I always seem to end up shopping for birthday or Christmas gifts at the last minute.
33. I usually buy even an essential item at the last minute.
34. I usually accomplish all the things I plan to do in a day.
35. I keep telling myself, "I'll do it tomorrow."
36. I usually take care of all the tasks I have to do before I settle down and relax for the evening.

Note. Reverse-keyed items: 1, 3, 8, 10, 11, 19, 20, 22, 24, 27, 29, 30, 31, 34, 36

List of Figures

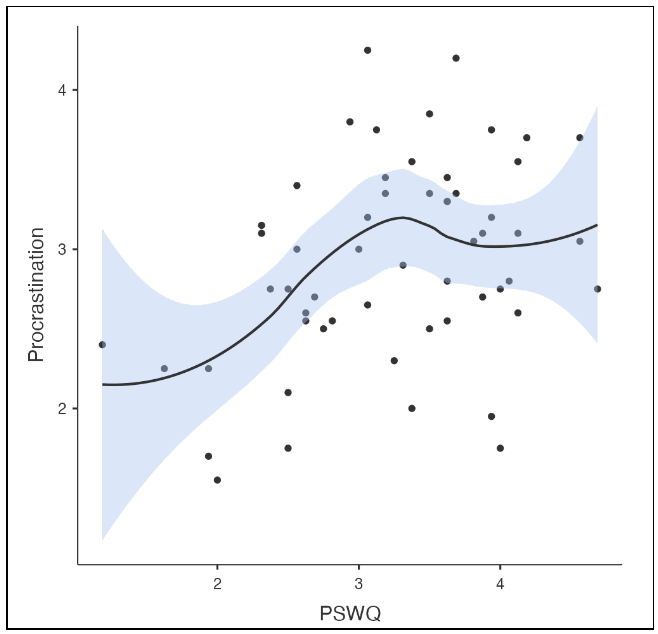

Figure 1. Scatter plot for correlation matrix.

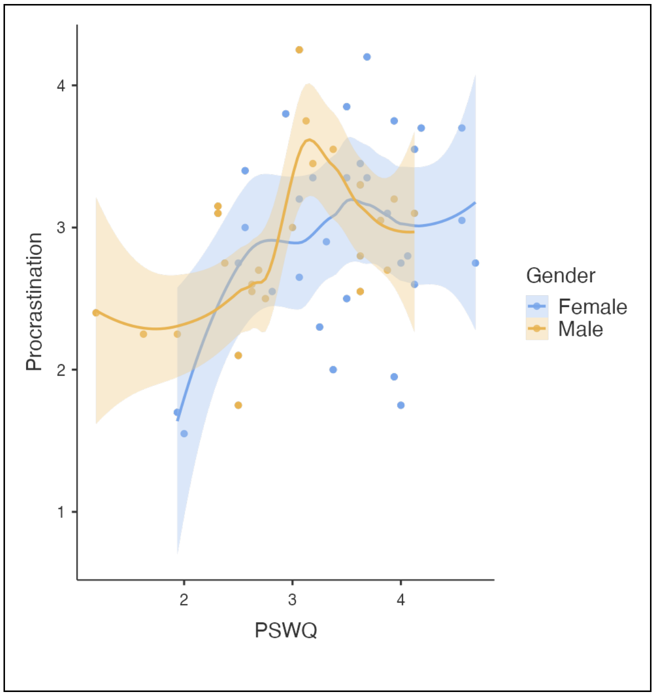

Figure 2. Scatter plot for descriptive statistics divided by gender.

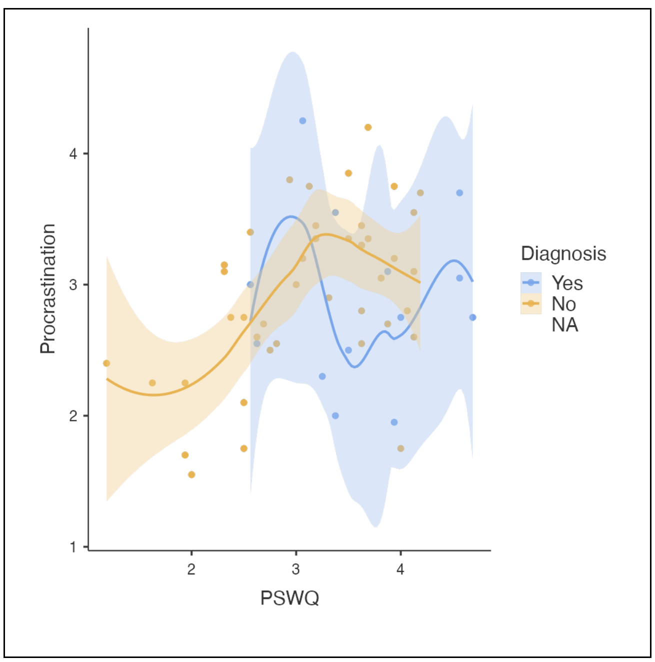

Figure 3. Scatter plot for descriptive statistics divided by history of anxiety diagnosis.

Note. Smooth continuous lines of both colours are regression lines corresponding to categories shown in the legend, shadings around them show standard error for all respective variables.

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Declaration on the use of generative Artificial Intelligence (AI) systems

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Research Paper – Introduction to Academic Working
Titel Prüfungsarbeit | Title of the exam

I have used the following artificial intelligence (AI)-based tools in the creation of my work:

1. ChatGPT
2. Claude

I further declare that

☑ I have actively informed myself about the performance and limitations of the above-mentioned AI tools,
☑ I have marked the passages taken from the above-mentioned AI tools,
☑ I have checked that the content generated with the help of the above-mentioned AI tools and adopted by me is actually accurate,
☑ I am aware that, as the author of this work, I am responsible for the information and statements made in it.

I have used the AI-based tools mentioned above as shown in the table below.

AI-based support tool	**Usage**	**Parts of the work affected**	**Remarks**
ChatGPT (GPT-5.5), OpenAI	Editing in-text citations, checking consistency between citations and the reference list	all chapters	
Claude (Sonnet 4.6), Anthropic PBC	Additional reference search	Introduction & Discussion	

Place, date, signature of the student

Berlin, 17.06.2026

НИКИТА АЛЕКСАНДРОВИЧ НЕВСКИЙ

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Declaration on the use of generative Artificial Intelligence (AI) systems

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Research Paper – Introduction to Academic Working
Titel Prüfungsarbeit | Title of the exam

I have used the following artificial intelligence (AI)-based tools in the creation of my work:
1. Grammarly

I further declare that

- ☑ I have actively informed myself about the performance and limitations of the above-mentioned AI tools,
- ☑ I have marked the passages taken from the above-mentioned AI tools,
- ☑ I have checked that the content generated with the help of the above-mentioned AI tools and adopted by me is actually accurate,
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I have used the AI-based tools mentioned above as shown in the table below.

AI-based support tool	**Usage**	**Parts of the work affected**	**Remarks**
Grammarly (free Version 1.2.99.1479), Grammarly	Spelling and grammar correction	all chapters	
Claude	Additional reference searches and streamlining the section flow	Discussion and possible limitations	

Place, date, signature of the student

Berlin, 17.06.2026

Felisha Ann Wade